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Asymmetric Relations Between Needs and Data Availability in Predictive Maintenance Systems

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Abstract

Maintenance decisions can remain consequential even when dedicated condition information is persistently weak or unavailable. This paper develops Limited-Data Predictive Maintenance (LD-PdM) as a conceptual characterization of that information and decision regime. Limited data refers here to the structure and relevance of the available evidence, rather than sample count alone. Four asymmetries organize the argument: weak condition observability, sparse and selective failure evidence, constraints on information acquisition, and the coexistence of uncertain individual states with possible population-level planning value. The framework connects proxy, contextual, historical, and prior evidence to explicit beliefs, decision consequences, and information-gathering choices. It distinguishes usage and exposure from degradation, event probability, and maintenance priority, and identifies assumptions required for each transition. A worked qualitative railway-station example traces a conditional inspection comparison between two doors and shows how changed routing evidence can invalidate it. Further scenarios address conflicting evidence, shared bias, and uninformative proxies. The contribution is an operational characterization and synthesis that connects established small-data prognostics and health management, virtual sensing, and probabilistic maintenance decision analysis to the feasible evidence structure. It provides criteria for identifying the regime, tracing an action rationale, and recognizing when an ordering must be withheld. The analysis is conceptual; predictive calibration and maintenance benefit remain questions for subsequent evaluation.

Keywords: predictive maintenance; limited data; partial observability; proxy information; Bayesian inference; uncertainty; risk-based maintenance; railway stations; inspection planning; decision support

1. Introduction

Predictive maintenance (PdM) connects evidence about an asset to decisions about its future operation, inspection, servicing, or replacement. Diagnosis and remaining useful life (RUL) prediction are important components of this process, but the operational purpose is to choose an appropriate action. Reviews describe the development of PdM, machine-learning methods, and their relationship to industrial information infrastructures [1–3]. The availability of informative measurements influences which parts of this workflow can be implemented. It does not, however, determine whether a maintenance decision has to be made.

Limited evidence is already a recognized problem in prognostics and health management (PHM). Li et al. [4] review limitations in monitoring quantity and quality, economic and operational barriers, and data-efficient learning. Their scope includes maintenance execution, multimodal records, physical knowledge, fleet information, and uncertainty.

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The question here is how to characterize maintenance settings in which feasible condition-sensitive channels remain weak and indirect evidence materially carries the reasoning. This characterization builds on the established account of scarcity and its persistent causes.

We call this regime *Limited-Data Predictive Maintenance* (LD-PdM). The term describes an evidence structure relative to a maintenance choice, rather than a threshold on dataset size. Consider a population of infrastructure devices whose maintenance affects service continuity, while the feasible information budget provides little dedicated condition monitoring for each device. Passenger flows, event logs, service records, documentation, and engineering knowledge may nevertheless inform hypotheses about usage and maintenance needs. The central asymmetry is that the consequence of a maintenance choice and the availability of informative evidence need not increase together. A useful characterization must explain both the opportunity to use such evidence and the limits of doing so.

This is also adjacent to established decision analysis. Srinivasan and Parlikad [5] assess condition-monitoring value for infrastructure maintenance using a partially observable Markov decision process (POMDP), while Memarzadeh and Pozzi [6] connect information value to inspection, monitoring, and system-level scheduling. Bayesian-network applications in safety, reliability, and risk assessment likewise provide established tools for integrating uncertain evidence [7]. LD-PdM uses these foundations to bring the feasible observation architecture to the foreground and organize a system-level account of maintenance reasoning from heterogeneous indirect evidence.

The paper makes three conceptual contributions. First, it gives operational inclusion criteria and boundary cases for LD-PdM, organized around four asymmetries in observability, failure evidence, acquisition costs, and scale. Second, it separates the inferential links from proxy evidence to usage, exposure, condition, and future events from the decision rule that determines an action. Third, it applies these distinctions to a worked qualitative station example in which an inspection ordering is both justified conditionally and withdrawn when its evidential basis changes. Together, these elements provide a way to identify the regime, trace what each source can support, and state the assumptions on which an action depends. The analysis is conceptual, and the station setting and evidence are hypothetical.

Section 2 defines the regime and its asymmetries. Section 3 describes evidence and belief formation, and Section 4 connects them to maintenance choices. Section 5 works through a hypothetical station example. Section 6 discusses overlap with existing approaches, limitations, and evaluation questions; Section 7 concludes.

2. The LD-PdM Regime and Its Asymmetries

2.1. Operational definition and boundary cases

LD-PdM is defined relative to an asset or population, a maintenance choice, and a planning horizon. It applies when the following conditions hold together:

(1) The maintenance choice has a consequential operational or economic outcome.
(2) A condition quantity relevant to that choice remains weakly informed by the feasible set of condition-sensitive channels because of persistent deployment, access, or acquisition constraints.
(3) Proxy, contextual, historical, or prior evidence therefore makes a material contribution to the assessment of future maintenance needs.

Here, persistent means that the limitation is expected to remain relevant over the decision horizon. It need not be permanent. The same asset can move between regimes if its instrumentation, data access, operating context, or maintenance objective changes.

The framework requires the limitations of the evidence to remain explicit when assessment is translated into action. This is a modelling requirement, distinct from the conditions identifying the regime: an overconfident implementation does not make its observation

Table 1. Qualitative boundary cases for LD-PdM. The categories can overlap; no numerical observability or cost threshold is implied.

Evidence setting	Maintenance implication	Relationship to LD-PdM
Few labelled samples, with an informative dedicated condition channel	Data-efficient learning may improve an existing assessment	Small sample count alone does not establish LD-PdM
Many operational records, with weak information about condition	Usage or context must be interpreted through explicit assumptions	LD-PdM can apply despite a large record volume
Dedicated monitoring and indirect evidence coexist	Different choices or failure mechanisms have different information needs	Mixed regimes are possible within one system
Additional monitoring is feasible and worthwhile for the choice	Acquiring condition information may change the observation regime	Weak observability should be reconsidered, not assumed permanent
Available evidence cannot distinguish relevant alternatives	A differentiated action requires another basis or further information	Naming the regime does not make inference informative

constraints disappear. Equally, a missing channel does not establish LD-PdM if other feasible channels already provide sufficient information for the choice. The asymmetry concerns the mismatch between consequential maintenance choices and the information available to support them; the four perspectives below explain how that mismatch arises and what follows from it.

These criteria concern information content, not simply the presence of sensors. A machine may produce abundant operational records while providing weak evidence about a particular degradation mechanism. Conversely, a small dataset may contain a strongly informative condition measurement. Moreover, degradation and RUL are inferred in many sensor-rich systems as well. The distinction is the availability and relevance of condition-sensitive channels, not whether a physical state is literally observed without a model. Table 1 illustrates the boundary.

2.2. Asymmetry of observability

The operational importance of a variable does not guarantee an informative measurement channel. A planner may need to anticipate loss of service while observing only traffic, location, occasional alarms, or service interventions. Such records can constrain hypotheses about device usage without identifying condition. Figure 1 separates the available evidence from the model-dependent quantities required for decisions.

Indirect measurement and virtual sensing offer relevant methodological analogies. Process control and engine monitoring provide examples of inference from secondary measurements [8,9], while infrastructure studies use vehicle or bogie responses to investigate bridge or rail properties [10,11]. These works demonstrate particular indirect-measurement formulations; they do not establish that passenger-flow data identify station-device degradation. LD-PdM requires each proposed proxy to be connected to the quantity it can actually inform.

2.3. Asymmetry of failure evidence

The evidence generated by operation and maintenance can differ from the evidence required to learn failure behaviour. Rare faults provide few examples, operating conditions differ, and intervention changes what is subsequently observed. A replacement before failure may leave a right-censored lifetime; periodic inspection may locate an event only within an interval. Missing work orders are a separate problem. A maintenance interven-

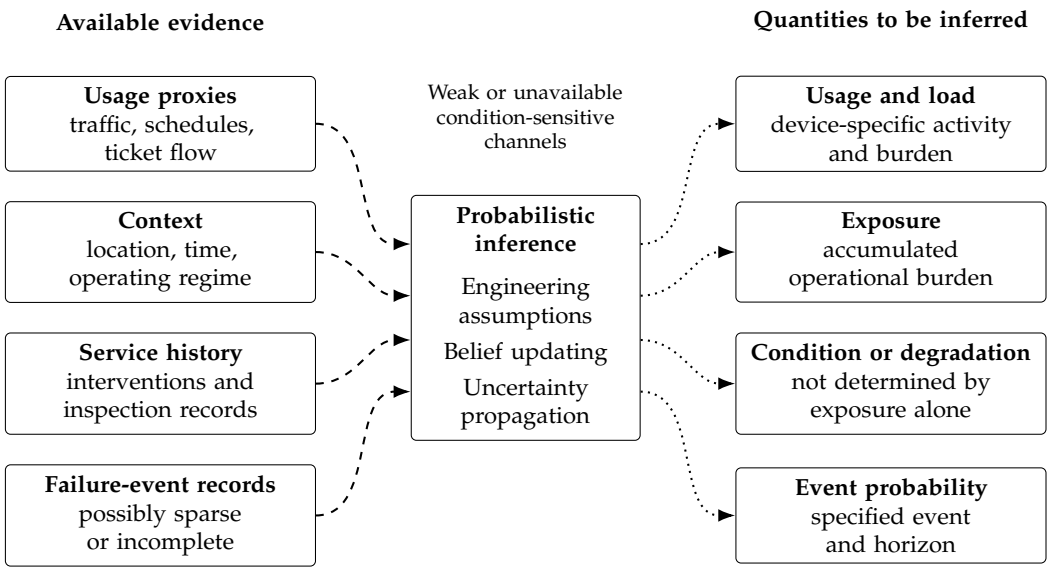


Figure 1. Conceptual separation between available evidence, probabilistic reasoning, and quantities used in maintenance decisions. Links represent model-dependent interpretation, not guaranteed state reconstruction. Weak condition-sensitive evidence can leave condition or event probabilities unresolved even when usage is partly informative.

tion also changes the physical state, so it cannot always be represented as an observation ending without consequence.

Statistical work on interval-censored data provides machinery for partially observed event times [12], but its applicability depends on the observation assumptions. In maintenance, inspections and replacement may be triggered by suspected deterioration, making the observation process informative. Accordingly, absence of a recorded failure should not automatically be interpreted as evidence of healthy condition. A conceptual framework must preserve the distinction between survival, intervention, and missing records; a later quantitative model must justify its treatment of each.

2.4. Asymmetry of information costs and decision consequences

The value of a maintenance decision and the cost of improving its evidence are different quantities. The latter includes acquisition and installation, integration with existing systems, commissioning and validation, connectivity and data processing, and the continuing operation and maintenance of the monitoring arrangement. The importance of deployment cost is also recognized in small-data PHM [4]. It motivates a conditional applicability statement: LD-PdM is relevant where better dedicated information cannot presently be obtained within the operator's feasible resources and constraints.

This condition must be assessed for the particular asset function and decision. Unit purchase price does not determine criticality, and an elevator or door may have substantial safety, accessibility, or service consequences. Additional monitoring can therefore be justified for a consequential function even in a population of ordinary devices. Equally, a station example does not establish that sensing is generally uneconomic for an asset class. Condition-monitoring value depends on what the information changes in the maintenance policy [5]; existing operational data also have integration and interpretation costs.

Table 2. Quantities distinguished in the conceptual framework. The final column identifies an inference limit, rather than a missing numerical parameter to be supplied in this paper.

Quantity	Meaning	Interpretive limit
Evidence	Proxy observations, context, service records, documentation	Record volume does not establish condition information
Usage and exposure	Device activity and accumulated operational burden	Passenger traffic is not a device-cycle counter or wear measurement
Condition	Physical degradation or functional state	Similar exposure does not imply equal condition
Event probability	Probability of a specified event within a declared horizon	A service threshold and a physical failure are different events
Risk	Consequences evaluated together with event uncertainty	An event probability alone does not specify consequences
Priority and action	A choice given beliefs, consequences, and constraints	Priority is a decision output, not a physical state

2.5. Asymmetry of scale

Uncertain individual states can coexist with useful population-level questions: where exposure concentrates, which groups merit inspection, or how limited attention should be distributed. Hierarchical modelling can share information across appropriately related units; Bayesian IoT failure modelling provides one example [13]. Risk-based and opportunistic maintenance studies illustrate the importance of system consequences and coordinated interventions [14,15]. These are established modelling opportunities, not a guarantee that a larger population makes weak evidence sufficient.

Pooling requires a defensible relationship between units, including their mechanisms, environments, and maintenance histories. Shared routing errors or a common misspecified proxy relationship do not disappear when more assets are included. Heterogeneous asset classes may require different models. Furthermore, estimating total burden and selecting the particular asset requiring action are different tasks: identical evidence and priors do not justify a data-derived individual ordering. The scale asymmetry is therefore an opportunity for decision support under stated assumptions, rather than an automatic reduction of uncertainty.

Taken together, the asymmetries distinguish the information architecture, the observation process, the acquisition constraint, and the level at which a choice is made. Weak relevant information and consequential choices are defining conditions. Rare events and population scale are important complications or opportunities, but need not occur in every LD-PdM setting. Their practical role is to determine what can be inferred, what remains assumed, and which decision claims would require additional evidence.

3. Evidence and Belief Formation

3.1. Separating observations from maintenance quantities

A proxy is informative only through a relationship with a specified target. Passenger counts might constrain use of an access route; a device-use model might translate that flow into operating cycles; cycles might inform cumulative exposure. None of these links, by itself, measures damage or establishes a failure probability. Age, maintenance quality, environment, design, and latent defects may alter the relationship between exposure and condition. Table 2 distinguishes the objects needed to describe that chain.

The evidence may combine operational aggregates, contextual variables, inspection and intervention histories, and engineering documentation. Their roles should remain identifiable. A timetable constrains possible demand; a service record describes an intervention; documentation supplies a mechanism or service requirement. Treating these as interchange-

able features obscures the observation assumptions. Context-dependent HVAC diagnostics and virtual-sensing studies provide examples of using alternative signals [16,17], while maintenance decision systems illustrate heterogeneous information integration [18]. Their particular data requirements should not be assumed to hold in an unsensored station.

3.2. An explicit belief state

Let $E_{1:t}$ denote the evidence available up to decision time t, and let M denote the assumed observation and evolution model. For asset i, a conceptual belief state can be written as

$$b_{i,t}(u, d, \theta) = p(U_{i,t} = u, D_{i,t} = d, \Theta = \theta \mid E_{1:t}, M), \tag{1}$$

where $U_{i,t}$ represents usage or accumulated exposure, $D_{i,t}$ physical condition, and Θ uncertain parameters connecting observations, use, and condition. This notation distinguishes evidence from model assumptions. It does not prescribe a distribution, assert that the variables are identifiable, or supply a fitted model. A defensible implementation would specify its likelihoods, evolution assumptions, and update procedure.

If a future event is required for a decision, it must be defined separately. For example, with T_i^F the time of a specified failure event and Δ the planning horizon,

$$q_{i,t}^F(a, \Delta) = \Pr\Big(t < T_i^F \leq t + \Delta \mid E_{1:t}, M, a\Big), \tag{2}$$

denotes a model-dependent event probability under the specified prospective action a, rather than an association conditional on an action selected by an earlier policy. Evaluating it requires a justified transition and event model; a belief about exposure is insufficient. The probability of reaching a service condition would have a different event definition. Failure consequences must then be supplied before either probability can be interpreted as risk.

Bayesian updating supplies established tools for learning model parameters from observations [19]. Bayesian networks can organize dependencies among uncertain inputs and states, and expert systems can make judgment-based assumptions visible [7,20]. The choice of representation should follow the evidence and decision requirements. A graph is useful for exposing assumed links; drawing an arrow does not validate the corresponding relationship. Likewise, a structured score represents a belief distribution only if its probabilistic meaning and uncertainty treatment are established.

3.3. Priors, dependence, and limits of identification

Engineering knowledge may constrain plausible routing, device-use mechanisms, deterioration sensitivities, or service conditions. Such information belongs in an explicit assumption register with its source and scope. Documentation, expert judgment, and previously observed behaviour need not have equal evidential strength. Multiple records derived from the same original traffic estimate are also not independent confirmations. A model should preserve their dependence rather than treating duplication as increased support.

When observations weakly constrain a parameter, its posterior can remain strongly dependent on the prior or chosen model class. Honest uncertainty representation therefore includes sensitivity to plausible alternative assumptions and identification of links that the available evidence cannot resolve. In LD-PdM, this can mean stopping at an exposure comparison rather than extending the chain to a numerical failure probability. Inspection or a targeted measurement may then be more useful than a more elaborate inference algorithm applied to the unchanged evidence.

4. Maintenance Choices under Weak Information

4.1. From beliefs to feasible actions

Beliefs become decision-relevant through their consequences for feasible actions. At a decision time, the planner should specify the asset function, planning horizon, candidate actions, and constraints. Depending on the task, these may include inspection, a predefined service intervention, or deferment within the applicable maintenance programme. Access windows, staffing, service continuity, and mandatory requirements restrict the feasible set. A ranking without an intended action or horizon remains an assessment product rather than a complete maintenance decision.

Established decision-theoretic notation expresses the connection as

$$a_t^\star \in \arg\min_{a \in \mathcal{A}_t} \mathbb{E}\big[L(a, S^a_{t:t+\Delta}) \mid E_{1:t}, M\big], \tag{3}$$

where $\mathcal{A}_t$ is the set of admissible choices, $S^a_{t:t+\Delta}$ denotes the uncertain system trajectory under choice a, and L represents the relevant consequences. These can include intervention effort, disruption, and adverse events. When a choice includes inspection, its evaluation must also include subsequent actions conditional on possible findings; an inspection is not valued as a terminal action with no use of its information. Equation 3 specifies what would make a belief useful for a choice; it is not an optimization algorithm or a calibrated objective for the station example. POMDP-based maintenance and information-value analyses develop this connection in quantitative settings [5,6].

For the qualitative example below, this logic becomes a conditional comparison. Suppose, for the mechanism considered, additional exposure makes an actionable inspection finding more plausible, while inspection burden, effectiveness, and the consequences of acting on the finding are comparable. An exposure difference could then support examining one asset first. This ordinal premise is an explicit assumption, not something established by the traffic proxy itself. It does not justify replacement or certify that the other asset is healthy. An action rationale should identify the link on which it depends and what additional finding would change the choice.

4.2. Inspection and the value of information

Inspection has an informational role as well as a resource cost. Its value depends on whether possible findings would alter a subsequent feasible action sufficiently to justify the inspection effort and disruption. Greater uncertainty can increase that opportunity, but it can also reflect a quantity irrelevant to the choice. Conversely, an asset with relatively concentrated beliefs can warrant intervention when the consequences and estimated state justify it. Uncertainty alone is therefore neither a universal reason to intervene nor a reason to defer.

Information gathering should be distinguished from changing the asset state. An inspection may update beliefs without repairing anything; servicing may change condition and the relevant future evidence. In a sequential problem, the expected consequences depend on later observations and actions as well as the present choice. Information-value work treats these dependencies explicitly [6]. The present framework uses this distinction to clarify possible actions without claiming to solve a full inspection and scheduling policy.

4.3. Population allocation and robustness

Across assets, prioritisation also requires a common objective and a resource constraint. Ranking by exposure, by expected consequence, and by expected inspection value can give different orders. Different device classes or service functions may therefore be incomparable under a single undifferentiated score. Coordinated maintenance and infrastructure

scheduling involve additional system constraints [14,15,21,22]; a proxy-based assessment supplies only part of that planning problem.

A robust qualitative recommendation states the circumstances under which its ordering persists and the circumstances under which it should be withdrawn. Relevant challenges include a changed traffic route, incorrect grouping of assets, inconsistent service histories, and a common bias affecting several proxy estimates. Stable rankings are not necessarily correct rankings. Where the evidence cannot distinguish the relevant choices, the appropriate output is an explicit unresolved comparison or a request for further information, rather than a more precise-looking priority number.

5. Worked Conceptual Example: A Railway Station

5.1. Setting and assumed evidence

Consider a hypothetical station with a concourse, platform access routes, automatic doors, elevators, and escalators. The example assumes that dedicated condition-sensitive channels are insufficient for the maintenance choice under consideration. It does not assume that instrumentation is absent from every device or economically unjustified for these asset classes. Consequences depend on function and alternatives: an elevator serving an otherwise inaccessible route can require different treatment from one with equivalent accessible alternatives. Operational importance and safety requirements must therefore be assessed for each function, rather than assigned to an entire device class.

For this example, suppose the planner has zone-level passenger counts or ticket-flow summaries, timetables and platform assignments, a station layout, device documentation, and dated inspection and service records. These are assumed inputs to a hypothetical example. The maintenance choice concerns inspection, service, or deferment between the current planning review and the next planned review. Existing mandatory service, safety, and accessibility requirements constrain the feasible actions; weak proxy evidence cannot relax them.

5.2. From passenger context to conditional beliefs

Passenger information first informs possible device use through an assumed routing relationship. Counts at an entrance do not identify the route taken by every passenger or the device selected. Timetables and platform assignments can constrain plausible routes, while closures and alternative access paths can change them. Routing uncertainty must remain attached to any resulting device-level usage belief. Ticket-flow and passenger-count summaries may also share an upstream source; their agreement is then weaker corroboration than agreement between independent observations.

Station topology gives this interpretation a system-level structure. A graph of entrances, circulation zones, platform connections, and accessible alternatives can encode which devices could serve a passenger stream without claiming to reconstruct individual trajectories. An outage or planned intervention may redirect that stream and change use of neighbouring assets. Device exposure is consequently conditional on route availability as well as demand. If the planner considers taking a device out of service, the resulting redistribution belongs to the consequences of that choice; the pre-intervention exposure map cannot simply be assumed to remain valid.

The next link is device-specific. For an automatic door, passage demand can inform an opening-cycle estimate only under assumptions about grouped passage, reopening, and control behaviour. For an elevator, passenger demand alone does not determine trips, occupancy, travel distance, or the distribution of loads. For an escalator, a continuously running operating regime differs from a demand-controlled regime: passenger flow and

Table 3. Assumption register for the hypothetical station. None of these relationships is estimated or validated in the present paper.

Inferential link	Assumption required	Evidence that could challenge it
Station flow to device use	Coverage, routing, and device-choice assumptions remain applicable during the planning interval.	Local passage observations, closures, or changed platform use.
Device use to exposure	The selected mechanism and operating mode explain the usage quantity being accumulated.	Cycle/run-time records, controller settings, or changed operating practice.
Exposure to condition	Relevant initial-state, degradation, and maintenance-effect assumptions are credible for this asset.	Inspection findings, component history, or signs inconsistent with the assumed mechanism.
Condition to future event	A specified event, horizon, and applicable conditional model are available.	Dated event/inspection records and checks of predictive calibration.
Beliefs to action	The exposure–finding relationship, inspection burden, consequences, and constraints justify the comparison.	Inspection findings, operator review, and access dependencies.

run time describe different aspects of use. Consequently, a common passenger indicator cannot be converted into a common wear score across these mechanisms.

Temporal structure matters for the same reason. Equal passenger totals spread evenly through the day or concentrated around train arrivals need not imply equal device activity. Where one opening serves several passengers, clustering can change the relation between passages and cycles. Where use is continuous, running duration may remain similar despite a change in passenger count. The relevant aggregation interval therefore follows the hypothesized mechanism and decision: a daily total can support a broad exposure discussion while concealing the timing needed to plan inspection without disrupting a concentrated demand period.

Usage may then inform a model of accumulated exposure over a stated interval, conditional on the mechanism being considered. Exposure is an intermediate description of operational burden, not an observation of damage. Connecting it to physical condition requires further assumptions about initial state, maintenance effects, environment, and the applicable degradation mechanism. A service record can modify those beliefs without establishing that the device was restored to an as-new state. Absence of a recorded fault is also compatible with incomplete reporting or earlier preventive intervention. Table 3 identifies where additional evidence could challenge the chain.

An event probability requires an additional, explicit event model. Reaching a documented service condition, discovering a defect at inspection, and losing required function before the next review are different events. The present example does not assign probabilities to any of them. The related door-and-elevator study develops Bayesian proxy-flow modelling and scheduling integration [23]; its service-condition probabilities should not be interpreted as calibrated physical-failure probabilities. Here the station provides a setting for examining the evidential links and their decision implications, rather than reproducing that technical implementation.

Figure 2 summarises the conditional structure. Its arrows identify interpretive dependencies; they do not establish that the available inputs identify every downstream quantity.

5.3. Inspection, service, and deferment

Consider two hypothetical automatic doors, A and B, serving alternative concourse-to-platform routes. With limited inspection capacity before the next review, the planner

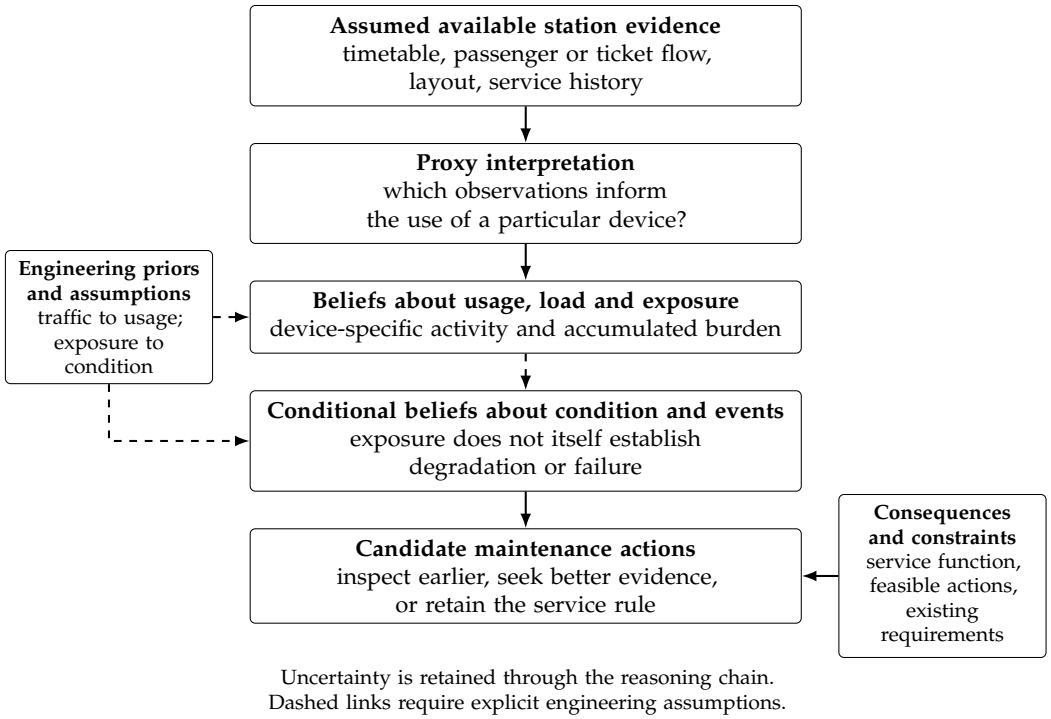


Figure 2. Schematic evidence-to-decision structure for the hypothetical station. Usage and exposure depend on proxy interpretation; condition and future-event beliefs require additional models and evidence. Assumptions and uncertainty apply throughout. The diagram contains no measured or estimated station outputs.

must choose which door to inspect first; no mandatory requirement determines the order. Assume comparable mechanisms, controller settings, baseline condition beliefs, inspection burdens, detectability of relevant findings, and benefits from acting on them. Dated service records inform these assumptions without establishing equal physical condition. Suppose zone counts, timetables, layout, and route availability support more passages through A under plausible route allocations. Further assume that passenger grouping and opening behaviour preserve greater cycle-related exposure for A; passenger totals alone cannot establish this contrast.

For the mechanism considered, assume greater exposure makes an actionable inspection finding more plausible and that using such a finding can improve the subsequent maintenance choice. Under the stated comparability assumptions, this supports inspecting A first. The recommendation identifies exposure as its evidential basis and the exposure–finding relationship as an additional premise. Servicing A would require an applicable service requirement or a relevant finding; the inspection ordering provides no assurance about B's condition. In an operational application, the assumed relationship would require checking against independent inspection evidence.

Now suppose a check of dated route assignments, corroborated by local observation, reveals that a platform reassignment was already in force during the exposure interval being compared. The allocation model omitted this change and attributed to A a stream that instead used an adjacent open passage. Under the revised plausible allocations, greater exposure for A is no longer supported. The planner therefore withdraws the inspection ordering and requests local usage evidence if a comparison is still needed. B does not automatically acquire priority. Mandatory service arrangements remain in force. The vignette demonstrates a conditional inspection comparison and its stopping rule. Extending this part of LD-PdM reasoning to a predictive maintenance policy would require a justified model of future condition or events and their action-dependent consequences.

Table 4. Five qualitative scenarios for the same planning interval. Every action remains subject to the applicable service requirements and functional consequences.

Evidence situation	Conditional reasoning and possible action	Limit on the conclusion
Consistent exposure contrast	Consider the more exposed door first under the stated exposure–finding premise and comparable inspection burden and consequences.	The premise requires independent checking; traffic alone establishes neither condition nor failure probability.
Similar expected exposure; unequal uncertainty	Clarify routing or inspect the less understood asset if resolving the uncertainty could change an important decision. Otherwise retain the applicable schedule.	Greater uncertainty alone supplies no intervention order.
Conflict or changed routing	Reconsider the mapping when counts, local observations, or a closure contradict its interpretation; seek local evidence before using the old ordering.	Conflicting sources should not be averaged into unwarranted confidence.
Common source of bias	Audit shared ticket coverage or route allocation before comparing affected groups; retain alternative routing interpretations.	Aggregation cannot remove a shared error. A uniform offset and a group-specific routing error affect comparisons differently.
Uninformative proxy	Retain the applicable service rule or seek more relevant evidence when counts cannot distinguish device use.	The framework supplies no exposure-based justification for differentiating these assets.

Direct service needs a separate rationale, such as an applicable service requirement or a relevant finding confirmed through inspection. Deferment means retaining the applicable service policy until the next review, subject to existing requirements and reconsideration when evidence changes. It is defensible only if that choice remains acceptable given the possible consequences and available alternatives. Neither a low traffic estimate nor a broad uncertainty interval establishes that deferment is safe. Equally, uncertainty alone does not establish that an intervention is beneficial: an inspection has decision value when its possible findings could change the choice enough to justify its burden.

Table 4 varies the evidential situation while keeping the distinction between information gathering and intervention explicit.

5.4. Population interpretation and scope

At station level, the planner can group tasks by access window or functional area while preserving these limits. Pooling usage evidence requires comparable mechanisms and a justified sharing structure; combining doors, elevators, and escalators merely because they share a zone does not supply one. Aggregated exposure may inform a workload discussion while leaving individual inspection order unresolved. Shared routing errors, reporting practices, and intervention histories can affect an entire group, so increasing the number of assets need not resolve the dominant uncertainty.

A spatial summary can support this discussion without becoming a map of physical damage. For example, the planner could distinguish a group whose exposure is plausibly concentrated on a busy access route from a group with uncertain allocation among alternative routes. This may help organize a local inspection visit or identify where a short counting exercise would clarify the evidence. The summary should retain both the assumed route structure and the origin of its uncertainty. A shaded zone labelled as a

hotspot would otherwise invite a stronger condition interpretation than the underlying information supports.

The example makes the proposed evidence-to-action reasoning explicit and identifies cases in which it must stop. Subsequent evaluation would need to test the proxy relationships against independent observations, examine their stability after operational changes, and compare decisions under the same resource and service constraints. The conceptual contribution specifies what such an evaluation would have to connect and distinguish.

5.5. Communicating the basis of a maintenance choice

The planner-facing output should distinguish a supported exposure contrast, a comparison sensitive to routing assumptions, and a comparison that the evidence cannot resolve. These are different reasons for attention even when presented in the same work list. An inspection request should state what is suspected, which evidence supports it, and what finding could change the next action. A request for better routing information should identify the disputed link rather than assigning the asset an unexplained high-uncertainty score. Communicating uncertainty in this form makes the model contestable by service personnel who can recognize a changed controller setting, an omitted intervention, or an inaccessible route. The assumption and scenario tables illustrate the content of such an output rather than prescribing a software interface.

6. Discussion

6.1. Relationship to small-data PHM

LD-PdM overlaps with the broad scope of small-data PHM. Li et al. [4] discuss maintenance execution in Section 1, uncertainty in Section 4.3.2.2, and multimodal and physics-informed approaches in Sections 6.2–6.3. Indirect evidence, prior knowledge, probabilistic treatment, fleet reasoning, and maintenance decisions are shared foundations of the present synthesis.

The difference is one of organizing emphasis. LD-PdM takes a specified maintenance choice and its feasible observation architecture as the starting point. The inclusion criteria identify when indirect evidence materially carries the reasoning; the quantity distinctions locate the assumptions needed at each inferential link; and the station scenarios show when those links support or defeat an inspection comparison. This organization lets a proposed action be examined through its evidence and assumptions before a learning or decision method is selected. Data-efficient learning can then operate within the characterization where its inputs and target are justified.

The perspectives overlap across applications. A setting with few condition samples may satisfy the LD-PdM criteria for one maintenance choice and fail them for another. Conversely, numerous passenger or operating records may remain weakly informative about the condition quantity relevant to a decision. Both perspectives require a justified relationship between available information and the target. The proposed synthesis traces that relationship through the system and the intended maintenance choice.

6.2. Relationship to other maintenance approaches

Virtual and indirect sensing contribute methods for inferring unavailable quantities from other observations. Their relevance depends on the particular measured quantity and model, as illustrated by the cross-domain examples in Section 2. LD-PdM places such estimates, their uncertainty, service histories, and context within a specified maintenance choice, making their contribution to the action rationale explicit.

Risk-based maintenance supplies consequence-aware objectives, while Bayesian inspection planning and POMDP formulations provide established ways to connect partial

observations, actions, and information acquisition [5,6,14]. The present contribution is the synthesis into explicit inclusion criteria, distinct inferential links, and a worked illustration for heterogeneous indirect evidence. Its value should be judged by whether this organization clarifies applicability and prevents unsupported transitions from available records to claimed maintenance knowledge.

Digital representations can organize asset identity, topology, histories, and beliefs without providing complete condition information. Digital-twin reviews and infrastructure decision frameworks offer relevant integration perspectives [24–26]. Such a representation is optional within LD-PdM. Its useful function is to retain the provenance of evidence and assumptions and connect them to the intended decision.

6.3. What a later evaluation should establish

The station example makes an inspection rationale traceable and shows which changed observation would withdraw it. Evaluation of predictive calibration, industrial savings, or superiority over an existing policy would require empirical observations or a declared simulation model and comparison design.

Future evaluation should separate the quality of proxy interpretation from the quality of the resulting choice. Where a quantity is observed for validation, useful checks include whether uncertainty estimates are calibrated and whether proxy relationships transfer across operating contexts. At the decision level, relevant outcomes include consequence-weighted missed events, unnecessary inspection or intervention, and the value of acquired information under equal resources. Ranking stability and predictive accuracy are useful supporting measures, but neither alone establishes decision benefit.

The reason is that a decision can depend on only part of the uncertainty about condition. Several plausible states may favour the same inspection, making exact discrimination among them unnecessary for that choice. Conversely, a small difference near a consequential action boundary may make additional information valuable. The appropriate question is therefore which unresolved distinctions could change an admissible action and its consequences. This explains why decision relevance can coexist with substantial state uncertainty, while also identifying when a coarse belief is insufficient. It is a conditional property of the decision problem, not an assertion that imprecise predictions are generally adequate.

A controlled synthetic experiment could isolate noise, shared bias, or uncertainty in the proxy-to-usage relationship. Its conclusions would remain conditional on the generated environment, and an exposure-based inspection objective would not establish failure reduction. Empirical work would additionally need observation and intervention records adequate to distinguish changing condition from changing recording practice. A calibrated failure model or industrial sensing break-even estimate therefore remains a separate research task.

6.4. Boundary conditions and failure of the proxy chain

The framework has several substantive limits. First, indirect evidence may be uninformative about the maintenance mechanism of interest even if it predicts usage. Second, different combinations of deterioration parameters and prior condition can explain the same records, leaving condition weakly identified. Third, an apparent population pattern can reflect common proxy error rather than shared physical burden. Explicit distributions help reveal these limitations but do not remove them.

Fourth, applicability is conditional on the acquisition constraint and the value of the choice. Better dedicated information should be considered when its feasible benefit justifies its lifecycle burden; a model is not justified simply because sensors are absent. If neither

the consequence of a decision nor the expected value of available evidence supports the modelling effort, LD-PdM provides a classification of the limitation rather than a reason to construct a new system. Safety, accessibility, and service obligations must be considered for the particular function, independently of unit price.

Finally, context and maintenance policy can change the observation process. Rerouting, refurbishment, different logging practices, or selective inspection can invalidate earlier proxy interpretations. An operational implementation would need a way to detect such changes, revise assumptions, and retain an audit trail. The research agenda is consequently as much about evidence quality, calibration, and communication as about more powerful inference methods.

7. Conclusions

Limited-Data Predictive Maintenance describes a regime in which consequential maintenance choices coexist with persistently weak feasible condition information, making indirect, contextual, historical, and prior evidence material to the reasoning. The asymmetry concerns the relationship between decision need and information availability. It is not defined by a small sample count alone, and it overlaps with established PHM, indirect sensing, and probabilistic maintenance decision analysis.

The operational criteria and quantity distinctions provide a way to examine a maintenance proposal: establish the information regime, identify what each source informs, and state the assumptions linking it to an action. The station vignette applies this structure to inspecting A first and withdrawing that ordering when routing evidence changes. Together with the boundary cases and scenario table, it shows how a maintenance rationale can remain explicit about both its basis and its stopping conditions. Predictive and operational performance remain questions for subsequent evaluation.

The practical implication is to match maintenance expectations to the available observation architecture while keeping the cost and potential benefit of better information open to reconsideration. Future work should evaluate proxy interpretation and maintenance consequences together, identify when shared assumptions dominate the result, and establish when additional inspection or monitoring is worthwhile.

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Abbreviations

PdM	Predictive maintenance
LD-PdM	Limited-Data Predictive Maintenance
PHM	Prognostics and health management
RUL	Remaining useful life
POMDP	Partially observable Markov decision process

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